

Description of the Excel file “BDT-September-2026”

The file contains several tabs:

- FictiveCalc – calculation of the “fictive” solid fraction, ψ_f , (eq 27a of the manuscript)
- Data – compilation of the brittle-ductile temperatures as functions of the strain rates for three polymers, PS, PMMA, and PVC, taken from Figure 5 of S. Wu, *J. Appl. Polym. Sci.*, 1976, **20**, 327–333
- CalculationsPMMA – parameters and calculations to compute the brittle-ductile transition strain rate of PMMA as a function of temperature, according to equation 22a of the manuscript
- PlotPMMA – graph of the calculated brittle-ductile transition temperature (BDT) as a function of the strain rate for PMMA: calculations (line) and data (squares).
- CalculationsPS – parameters and calculations to compute the brittle-ductile transition strain rate of PS as a function of temperature, according to equation 22a of the manuscript
- PlotPS – graph of the calculated brittle-ductile transition temperature (BDT) as a function of the strain rate for PS: calculations (line) and data (squares).
- CalculationsPVC – parameters and calculations to compute the brittle-ductile transition strain rate of PS as a function of temperature, according to equation 22a of the manuscript
- PlotPVC – graph of the calculated brittle-ductile transition temperature (BDT) as a function of the strain rate for PS: calculations (line) and data (squares).

All the calculations hinge on inverting eqs 27a-b to solve for T_f as a function of t_a . In the tab

FictiveCalc, we compute $Y = \ln(t_a / \tau_{el})$ vs. $Z = \frac{T_x}{T_g} \frac{T_g}{T_f}$ (combining eqs 27a and 27b). We

then fit the function $Z(Y)$ with a quadratic polynomial, and use this polynomial in the calculations for the three polymers.

In the tabs CalculationsPMMA, CalculationsPS, and CalculationsPVC, all parameters are introduced as discussed in the manuscript; the only adjustable parameter is X (highlighted in yellow). The experimental data for BDT vs. strain rate are fitted to the form, $T_{BD} = a \ln(S) + b$ (column Z) and the GRG nonlinear solver is used to minimize the χ^2 function in the strain rate range between 10^{-5} and 10^{-1} s^{-1} .

The tabs PlotPMMA, PlotPS, and PlotPVC show the comparison between the model fits and the experimental data of Wu. Those correspond to Figures 6, 7, and 8, respectively.